



Automated Highway Lane Following System using MATLAB

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Introduction

An Automated Highway Lane Following System utilizing MATLAB specially for Indian traffic system.

◆ Problem Statement:

A robust automated system is needed for accurate lane detection and vehicle stability for Indian roads that follows India traffic rules without human intervention.

◆ Motivation:

- **Reducing road accidents caused by lane departure.**
- **Enhancing autonomous vehicle navigation.**
- **Improving traffic flow and driving comfort.**

Novelty

- Adaptive vehicle positioning following Indian traffic rules (right-side overtaking)
- Adaptive vehicle positioning with minimal computational load
- Enhanced safety through continuous lane deviation alerts

Background Study

- Recent studies emphasize advanced algorithms for vehicle path planning, including adaptive control strategies to ensure precise lane following on highways.
- Traditional methods rely on Canny edge detection for lane identification.
- Most autonomous systems are designed for left-hand drive countries (overtaking from the left). Our system is customized for right-side overtaking, aligning with Indian traffic norms.
- MATLAB's Simulink and other toolboxes facilitates the validation of automated driving systems under different driving scenarios, ensuring reliable performance before real-world deployment.

Methodology

1. Program highway lanes and vehicles

- Uses Simulink and MATLAB's Driving Scenario Designer to model urban and rural roads.
- Runs real-time simulations with sensor data, AI-driven decision-making, and vehicle control.

2. Perception Algorithm

- Uses camera, radar, and LiDAR for environmental data collection.
- Detects lanes, vehicles, traffic signals, and road signs via MATLAB's Vision Detector Block.
- Applies computer vision techniques (e.g., laneMarkerDetector, traffic sign recognition).

3. Sensor Fusion

- Integrates multi-sensor data to enhance object tracking and reduce false detections.
- Utilizes MATLAB's Forward Vehicle Sensor Fusion for better decision-making.

3. Planning Algorithm

- Computes vehicle trajectory using path planning techniques like Dijkstra's, and Model Predictive Control (MPC).
- Optimizes lane changes based on Indian overtaking rules (right-side overtaking).

4. Control Algorithm

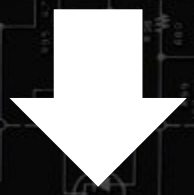
- Implements Model Predictive Control (MPC) for steering and speed adjustments.
- Ensures smooth lateral and longitudinal control to maintain lane discipline and obey speed limits.

5. Simulation & Testing

- Uses Simulink and MATLAB's Driving Scenario Designer to model urban and rural roads.
- Runs real-time simulations with sensor data, AI-driven decision-making, and vehicle control.

Present Status of Work

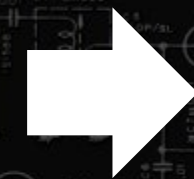
1. Two vehicles running with same speed on a Straight 3 lane track



2. Three vehicles running with same speed on a Circular 3 lane track



3. Five vehicles with different speeds running on a circular 5 lane track with five different trajectories results in collision and overtaking



4. A vehicle running in the left lane of a straight road overtakes only one front vehicle from the right side and comes back to its initial lane resulting in collision

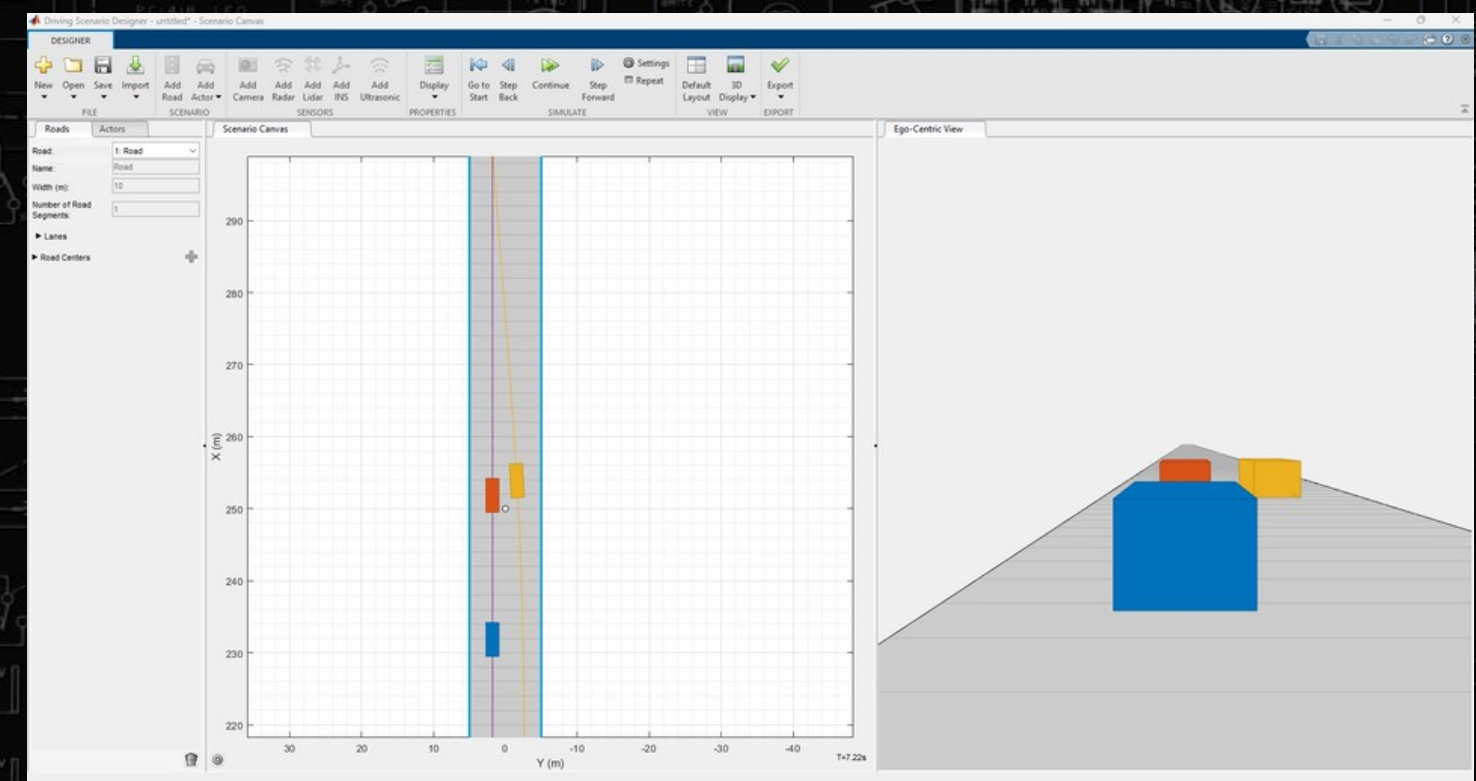
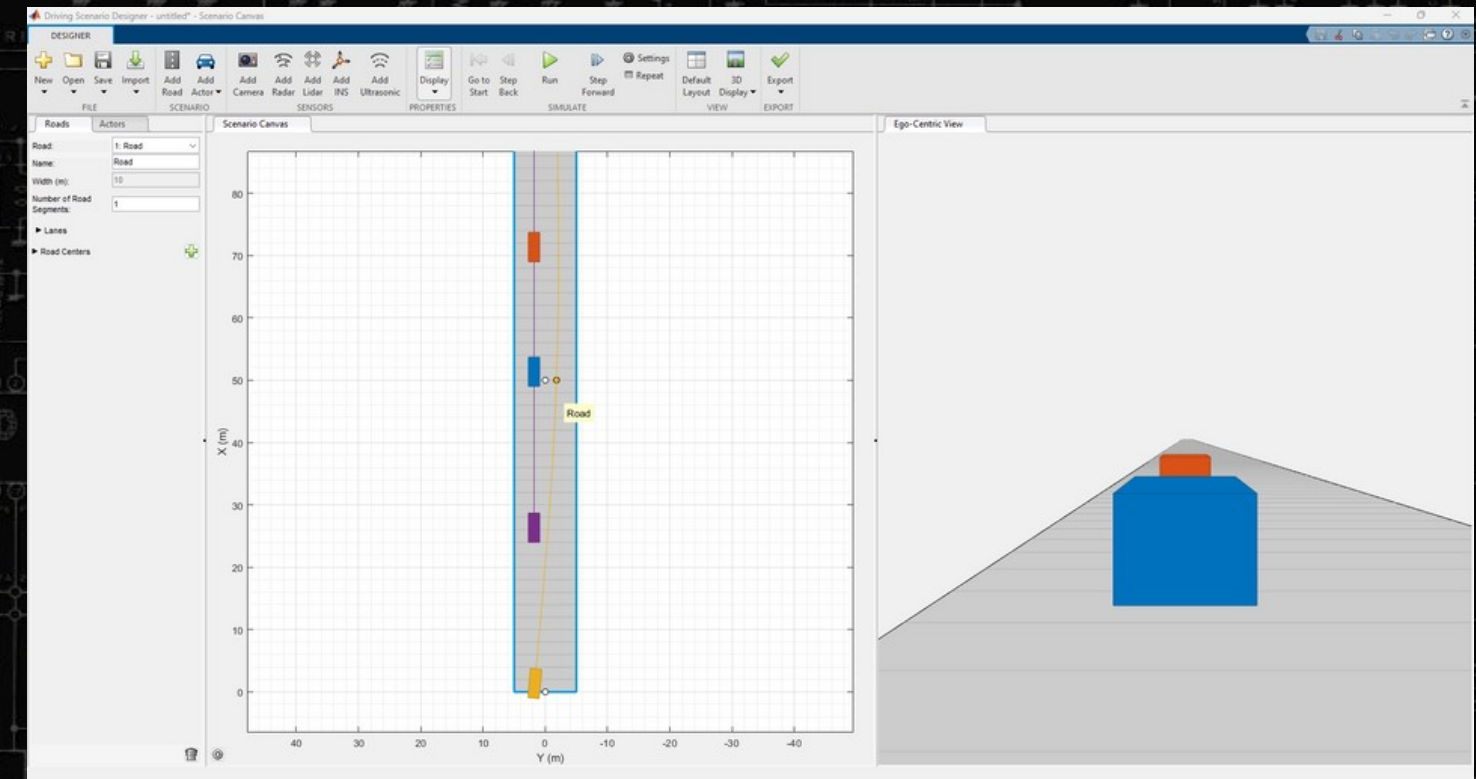
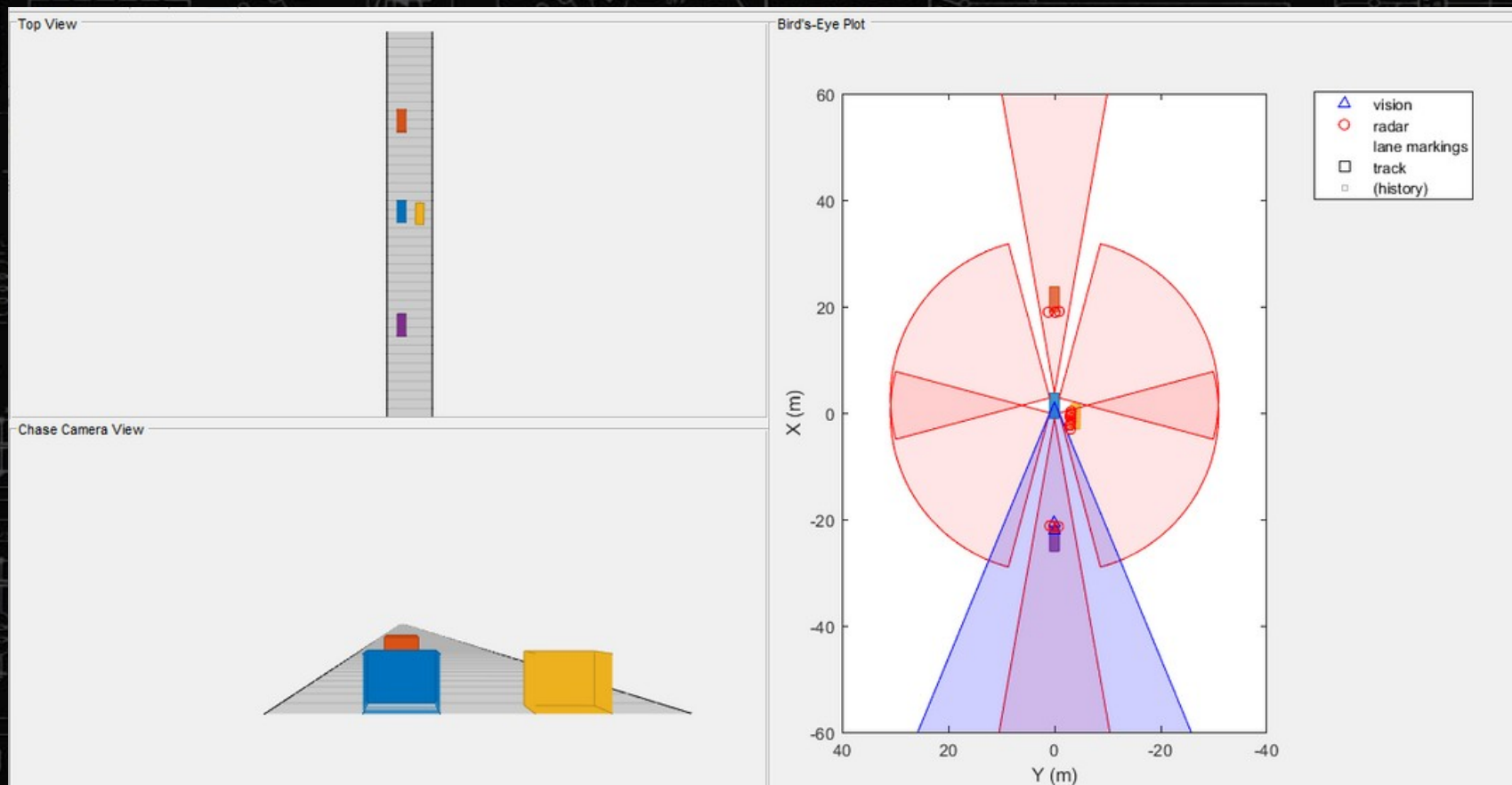


5. A vehicle running in the left lane of a straight road overtakes all the front vehicles from the right side and comes back to its initial lane



Result and Analysis

Three vehicles with same speeds running on the left lane of a straight 2 lane track. The passing vehicle was running in the left lane initially, after detecting the front vehicle it changed its trajectory to move to the right lane and comes back to the left lane after overtaking all vehicles.



Challenges and Remedies

Figuring out how things work – MATLAB is very much useful but also very much chaotic for beginners, we couldn't figure out what things comes first and how to code and connect it with other components as described in the methodology.

Remedies - We did a lot of experiments on our own and are doing a lot still.

Logic building for overtaking – We are facing challenges on building the logic how will a car overtake the front vehicle just like a skilled driver does. Our simulated car is currently going through the front vehicle instead of overtaking.

Remedies - We are reverse engineering the examples given by MATLAB and prebuilt driving systems available online.

Indian Traffic Complexity – Irregular driving, diverse traffic, and unpredictable roads made automation difficulty. So organizing a road according to real world is a challenge.

Remedies - We are going through MATLAB documentation on how to handle it.

Roadrunner Access – As roadrunner is not in our student license, we can not create 3D animation of our simulation.

Remedies - So we have planned to request for a 30 day trial of roadrunner after our simulation is complete.

Overall Progress

Timeline

Status: Not Planning ▾

Progress Bar ▾

» November 2024 mber

January

February

Open in Calendar

Quarter ▾

< Today >

25

2

9

16

23

30

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27

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Planning

+ New

Figuring out Driving Scenario Designer

Programming the vehicle parameters

Simulating scenarios manipulating vehicle parameters

Implementing Sensor Fusion

Simulation & Testing

Future Work

Improvement of Sensor Accuracy: Discuss the possibility of using more advanced sensors or higher-resolution cameras for better traffic monitoring.

Real-World Testing: Suggest deploying the system in real-world Indian traffic for more rigorous testing and refinement.

Machine Learning Enhancements: Explore using more advanced machine learning models, such as deep learning for better perception, decision-making, or even predicting traffic patterns.

Traffic Simulation: Build more detailed simulations of urban and rural Indian traffic to test the system under various conditions.

Collaboration with Industry: Propose working with local authorities, transportation agencies, or vehicle manufacturers to integrate your system with real-world vehicles and infrastructure.

References

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**Thank
You**